

## Independence of Path

we saw that when a real function  $f$  possesses an elementary antiderivative, that is, a function  $F$  for which  $F'(x) = f(x)$ , a definite integral can be evaluated by the fundamental theorem of calculus:

$$\int_a^b f(x) dx = F(b) - F(a). \quad (1)$$

Note that  $\int_a^b f(x) dx$  depends only on the numbers  $a$  and  $b$  at the initial and terminal points of the interval of integration. In contrast, the value of a real line integral  $\int_C P dx + Q dy$  generally depends on the curve  $C$ . However, there exist integrals  $\int_C P dx + Q dy$  whose value depends only on the initial point  $A$  and terminal point  $B$  of the curve  $C$ , and not on  $C$  itself. In this case we say that the line integral is **independent of the path**. For example,  $\int_C y dx + x dy$  is independent of the path.

A line integral that is independent of the path can be evaluated in a manner similar to (1). It seems natural then to ask:

*Is there a complex version of the fundamental theorem of calculus?*

*Can a contour integral  $\int_C f(z) dz$  be independent of the path?*

In this section we will see that the answer to both of these questions is yes.

**Path Independence** The definition of path independence for a contour integral  $\int_C f(z) dz$  is essentially the same as for a real line integral  $\int_C P dx + Q dy$ .

### **Definition** Independence of the Path

Let  $z_0$  and  $z_1$  be points in a domain  $D$ . A contour integral  $\int_C f(z) dz$  is said to be independent of the path if its value is the same for all contours  $C$  in  $D$  with initial point  $z_0$  and terminal point  $z_1$ .

At the end of the preceding section we noted that the Cauchy-Goursat theorem also holds for closed contours, not just simple closed contours, in a simply connected domain  $D$ . Now suppose, as shown in Figure 1, that  $C$  and  $C_1$  are two contours lying entirely in a simply connected domain  $D$  and both with initial point  $z_0$  and terminal point  $z_1$ . Note that  $C$  joined with the opposite curve  $-C_1$  forms a closed contour. Thus, if  $f$  is analytic in  $D$ , it follows from the Cauchy-Goursat theorem that

$$\int_C f(z) dz + \int_{-C_1} f(z) dz = 0. \quad (2)$$

But (2) is equivalent to

$$\int_C f(z) dz = \int_{C_1} f(z) dz. \quad (3)$$

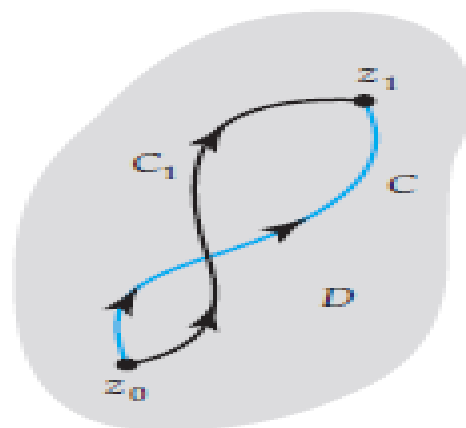


Figure 1 If  $f$  is analytic in  $D$ , integrals on  $C$  and  $C_1$  are equal.

## Theorem Analyticity Implies Path Independence

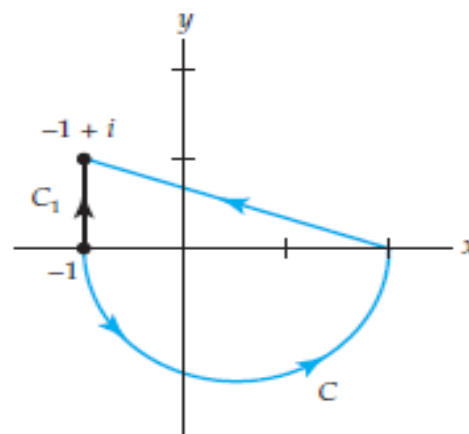
Suppose that a function  $f$  is analytic in a simply connected domain  $D$  and  $C$  is any contour in  $D$ . Then  $\int_C f(z) dz$  is independent of the path  $C$ .

### EXAMPLE 1 Choosing a Different Path

Evaluate  $\int_C 2z dz$ , where  $C$  is the contour shown in color in Figure 1.

**Solution** Since the function  $f(z) = 2z$  is entire, we can, in view of Theorem 1, replace the piecewise smooth path  $C$  by any convenient contour  $C_1$  joining  $z_0 = -1$  and  $z_1 = -1 + i$ . Specifically, if we choose the contour  $C_1$  to be the vertical line segment  $x = -1$ ,  $0 \leq y \leq 1$ , shown in black in Figure 1, then  $z = -1 + iy$ ,  $dz = i dy$ . Therefore,

$$\int_C 2z dz = \int_{C_1} 2z dz = -2 \int_0^1 y dy - 2i \int_0^1 dy = -1 - 2i.$$



A contour integral  $\int_C f(z) dz$  that is independent of the path  $C$  is usually written  $\int_{z_0}^{z_1} f(z) dz$ , where  $z_0$  and  $z_1$  are the initial and terminal points of  $C$ . Hence, in Example 1 we can write  $\int_{-1}^{-1+i} 2z dz$ .

There is an easier way to evaluate the contour integral in Example 1, but before proceeding we need another definition.

## Definition Antiderivative

Suppose that a function  $f$  is continuous on a domain  $D$ . If there exists a function  $F$  such that  $F'(z) = f(z)$  for each  $z$  in  $D$ , then  $F$  is called an antiderivative of  $f$ .

For example, the function  $F(z) = -\cos z$  is an antiderivative of  $f(z) = \sin z$  since  $F'(z) = \sin z$ . As in calculus of a real variable, the most general antiderivative, or indefinite integral, of a function  $f(z)$  is written  $\int f(z) dz = F(z) + C$ , where  $F'(z) = f(z)$  and  $C$  is some complex constant. For example,  $\int \sin z dz = -\cos z + C$ .

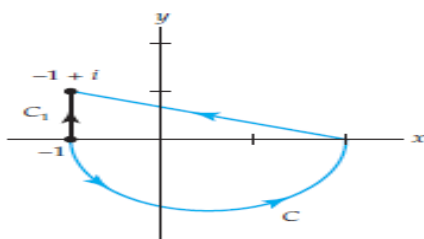
Since an antiderivative  $F$  of a function  $f$  has a derivative at each point in a domain  $D$ , it is necessarily analytic and hence continuous at each point in  $D$ .

We are now in a position to prove the complex analogue of (1).

**Recall, differentiability implies continuity.**

## Theorem ■ Fundamental Theorem for Contour Integrals

Suppose that a function  $f$  is continuous on a domain  $D$  and  $F$  is an antiderivative of  $f$  in  $D$ . Then for any contour  $C$  in  $D$  with initial point  $z_0$  and terminal point  $z_1$ ,



$$\int_C f(z) dz = F(z_1) - F(z_0). \quad (4)$$

### EXAMPLE 2 Applying Theorem

In Example 1 we saw that the integral  $\int_C 2z dz$ , where  $C$  is shown in Figure 1, is independent of the path. Now since the  $f(z) = 2z$  is an entire function, it is continuous. Moreover,  $F(z) = z^2$  is an antiderivative of  $f$  since  $F'(z) = 2z = f(z)$ . Hence, by (4) of Theorem 3.1 we have

$$\int_{-1}^{-1+i} 2z dz = z^2 \Big|_{-1}^{-1+i} = (-1+i)^2 - (-1)^2 = -1 - 2i.$$

### EXAMPLE 3 Applying Theorem

Evaluate  $\int_C \cos z \, dz$ , where  $C$  is any contour with initial point  $z_0 = 0$  and terminal point  $z_1 = 2 + i$ .

**Solution**  $F(z) = \sin z$  is an antiderivative of  $f(z) = \cos z$  since  $F'(z) = \cos z = f(z)$ . Therefore, from (4) we have

$$\int_C \cos z \, dz = \int_0^{2+i} \cos z \, dz = \sin z \Big|_0^{2+i} = \sin(2+i) - \sin 0 = \sin(2+i).$$

If we desire a complex number of the form  $a + ib$  for an answer, we can use  $\sin(2+i) \approx 1.4031 - 0.4891i$  Hence,

$$\int_C \cos z \, dz = \int_0^{2+i} \cos z \, dz \approx 1.4031 - 0.4891i.$$



## Some Conclusions

We can draw several immediate conclusions from Theorem 1. First, observe that if the contour  $C$  is closed, then  $z_0 = z_1$  and, consequently,

Previous theorem-

Fundamental theorem

for counter int.

$$\oint_C f(z) dz = 0. \quad (5)$$

Next, since the value of  $\int_C f(z) dz$  depends only on the points  $z_0$  and  $z_1$ , this value is the same for any contour  $C$  in  $D$  connecting these points. In other words:

*If a continuous function  $f$  has an antiderivative  $F$  in  $D$ , then*

$$\int_C f(z) dz \text{ is independent of the path.} \quad (6)$$

In addition, we have the following sufficient condition for the existence of an antiderivative.

*If  $f$  is continuous and  $\int_C f(z) dz$  is independent of the path  $C$  in a domain  $D$ , then  $f$  has an antiderivative everywhere in  $D$ .*

$$(7)$$

**Proof of (7)** Assume  $f$  is continuous and  $\int_C f(z) dz$  is independent of the path in a domain  $D$  and that  $F$  is a function defined by  $F(z) = \int_{z_0}^z f(s) ds$ , where  $s$  denotes a complex variable,  $z_0$  is a fixed point in  $D$ , and  $z$  represents any point in  $D$ . We wish to show that  $F'(z) = f(z)$ , that is,

$$F(z) = \int_{z_0}^z f(s) ds \quad (8)$$

is an antiderivative of  $f$  in  $D$ . Now,

$$F(z + \Delta z) - F(z) = \int_{z_0}^{z+\Delta z} f(s) ds - \int_{z_0}^z f(s) ds = \int_z^{z+\Delta z} f(s) ds. \quad (9)$$

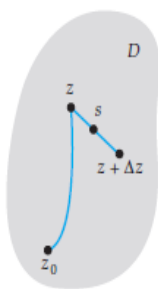
Because  $D$  is a domain, we can choose  $\Delta z$  so that  $z + \Delta z$  is in  $D$ . Moreover,  $z$  and  $z + \Delta z$  can be joined by a straight segment as shown in Figure . This is the contour we use in the last integral in (9). With  $z$  fixed, we can write

$$f(z)\Delta z = f(z) \int_z^{z+\Delta z} ds = \int_z^{z+\Delta z} f(z) ds \quad \text{or} \quad f(z) = \frac{1}{\Delta z} \int_z^{z+\Delta z} f(z) ds. \quad (10)$$

From (9) and the last result in (10) we have

$$\frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) = \frac{1}{\Delta z} \int_z^{z+\Delta z} [f(s) - f(z)] ds.$$

Now  $f$  is continuous at the point  $z$ . This means for any  $\varepsilon > 0$  there exists a  $\delta > 0$  so that  $|f(s) - f(z)| < \varepsilon$  whenever  $|s - z| < \delta$ . Consequently, if we choose  $\Delta z$  so that  $|\Delta z| < \delta$ , it follows from the *ML*-inequality that



$$\begin{aligned} \left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| &= \left| \frac{1}{\Delta z} \int_z^{z+\Delta z} [f(s) - f(z)] ds \right| \\ &= \left| \frac{1}{\Delta z} \right| \left| \int_z^{z+\Delta z} [f(s) - f(z)] ds \right| \leq \left| \frac{1}{\Delta z} \right| \varepsilon |\Delta z| = \varepsilon. \end{aligned}$$

Hence, we have shown that

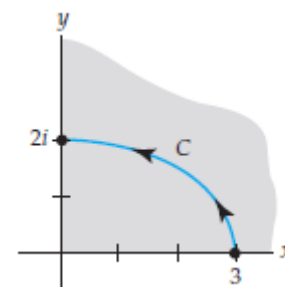
$$\lim_{\Delta z \rightarrow 0} \frac{F(z + \Delta z) - F(z)}{\Delta z} = f(z) \quad \text{or} \quad F'(z) = f(z).$$

If  $f$  is an analytic function in a simply connected domain  $D$ , it is necessarily continuous throughout  $D$ . This fact, when put together with the results in Theorem      and (7), leads to a theorem which states that an analytic function possesses an analytic antiderivative.

**Theorem**      **Analyticity Implies Path Independence**

### EXAMPLE 4 Using the Logarithmic Function

Evaluate  $\int_C \frac{1}{z} dz$ , where  $C$  is the contour shown in Figure 1.



**Solution** Suppose that  $D$  is the simply connected domain defined by  $x > 0$ ,  $y > 0$ , in other words,  $D$  is the first quadrant in the  $z$ -plane. In this case,  $\text{Ln } z$  is an antiderivative of  $1/z$  since both these functions are analytic in  $D$ . Hence by (4),

$$\int_3^{2i} \frac{1}{z} dz = \text{Ln } z \Big|_3^{2i} = \text{Ln } 2i - \text{Ln } 3.$$

$$\text{Ln } 2i = \log_e 2 + \frac{\pi}{2}i \quad \text{and} \quad \text{Ln } 3 = \log_e 3 \quad \log_e 2 - \log_e 3 = \log_e \frac{2}{3}$$

$$\int_3^{2i} \frac{1}{z} dz = \log_e \frac{2}{3} + \frac{\pi}{2}i \approx -0.4055 + 1.5708i.$$

**EXAMPLE 5** Using an Antiderivative of  $z^{-1/2}$

Evaluate  $\int_C \frac{1}{z^{1/2}} dz$ , where  $C$  is the line segment between  $z_0 = i$  and  $z_1 = 9$ .

**Solution** Throughout we take  $f_1(z) = z^{1/2}$  to be the principal branch of the square root function. In the domain  $|z| > 0$ ,  $-\pi < \arg(z) < \pi$ , the function  $f_1(z) = 1/z^{1/2} = z^{-1/2}$  is analytic and possesses the antiderivative  $F(z) = 2z^{1/2}$ . Hence,

$$\begin{aligned}\int_i^9 \frac{1}{z^{1/2}} dz &= 2z^{1/2} \Big|_i^9 = 2 \left[ 3 - \overbrace{\left( \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right)} \right] \\ &= (6 - \sqrt{2}) - i\sqrt{2}.\end{aligned}$$

# Cauchy's Integral Formulas and Their Consequences

In the last two sections we saw the importance of the Cauchy-Goursat theorem in the evaluation of contour integrals. In this section we are going to examine several more consequences of the Cauchy-Goursat theorem. Unquestionably, the most significant of these is the following result:

*The value of an analytic function  $f$  at any point  $z_0$  in a simply connected domain can be represented by a contour integral.*

After establishing this proposition we shall use it to further show that:

*An analytic function  $f$  in a simply connected domain possesses derivatives of all orders.*

# Cauchy's Two Integral Formulas

**First Formula** If  $f$  is analytic in a simply connected domain  $D$  and  $z_0$  is any point in  $D$ , the quotient  $f(z)/(z - z_0)$  is not defined at  $z_0$  and hence is *not* analytic in  $D$ . Therefore, we *cannot* conclude that the integral of  $f(z)/(z - z_0)$  around a simple closed contour  $C$  that contains  $z_0$  is zero by the Cauchy-Goursat theorem. Indeed, as we shall now see, the integral of  $f(z)/(z - z_0)$  around  $C$  has the value  $2\pi i f(z_0)$ . The first of two remarkable formulas is known simply as the **Cauchy integral formula**.

## **Theorem** **Cauchy's Integral Formula**

Suppose that  $f$  is analytic in a simply connected domain  $D$  and  $C$  is any simple closed contour lying entirely within  $D$ . Then for any point  $z_0$  within  $C$ ,

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz. \quad (1)$$

**Proof** Let  $D$  be a simply connected domain,  $C$  a simple closed contour in  $D$ , and  $z_0$  an interior point of  $C$ . In addition, let  $C_1$  be a circle centered at  $z_0$  with radius small enough so that  $C_1$  lies within the interior of  $C$ . By the principle of deformation of contours, we can write

$$\oint_C \frac{f(z)}{z - z_0} dz = \oint_{C_1} \frac{f(z)}{z - z_0} dz. \quad (2)$$

We wish to show that the value of the integral on the right is  $2\pi i f(z_0)$ . To this end we add and subtract the constant  $f(z_0)$  in the numerator of the integrand,

$$\begin{aligned} \oint_{C_1} \frac{f(z)}{z - z_0} dz &= \oint_{C_1} \frac{f(z_0) - f(z_0) + f(z)}{z - z_0} dz \\ &= f(z_0) \oint_{C_1} \frac{1}{z - z_0} dz + \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0} dz. \end{aligned} \quad (3)$$

we know that

$$\oint_{C_1} \frac{1}{z - z_0} dz = 2\pi i \quad (4)$$



and so (3) becomes

$$\oint_{C_1} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0) + \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0} dz. \quad (5)$$

Since  $f$  is continuous at  $z_0$ , we know that for any arbitrarily small  $\varepsilon > 0$  there exists a  $\delta > 0$  such that  $|f(z) - f(z_0)| < \varepsilon$  whenever  $|z - z_0| < \delta$ . In particular, if we choose the circle  $C_1$  to be  $|z - z_0| = \frac{1}{2}\delta < \delta$ , then by the *ML-inequality* the absolute value of the integral on the right side of the equality in (5) satisfies

$$\left| \oint_{C_1} \frac{f(z) - f(z_0)}{z - z_0} dz \right| \leq \frac{\varepsilon}{\delta/2} 2\pi \left( \frac{\delta}{2} \right) = 2\pi\varepsilon.$$

In other words, the absolute value of the integral can be made arbitrarily small by taking the radius of the circle  $C_1$  to be sufficiently small. This can happen only if the integral is 0. Thus (5) is  $\oint_{C_1} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$ . The theorem is proved by dividing both sides of the last result by  $2\pi i$ .

Because the symbol  $z$  represents a point on the contour  $C$ , (1) indicates that *the values of an analytic function  $f$  at points  $z_0$  inside a simple closed contour  $C$  are determined by the values of  $f$  on the contour  $C$ .*

*If  $f$  is analytic at all points within and on a simple closed contour  $C$ , and  $z_0$  is any point interior to  $C$ , then*

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz.$$

### EXAMPLE 1 Using Cauchy's Integral Formula

Evaluate  $\oint_C \frac{z^2 - 4z + 4}{z + i} dz$ , where  $C$  is the circle  $|z| = 2$ .

**Solution** First, we identify  $f(z) = z^2 - 4z + 4$  and  $z_0 = -i$  as a point within the circle  $C$ . Next, we observe that  $f$  is analytic at all points within and on the contour  $C$ . Thus, by the Cauchy integral formula (1) we obtain

$$\oint_C \frac{z^2 - 4z + 4}{z + i} dz = 2\pi i f(-i) = 2\pi i(3 + 4i) = \pi(-8 + 6i).$$

## EXAMPLE 2 Using Cauchy's Integral Formula

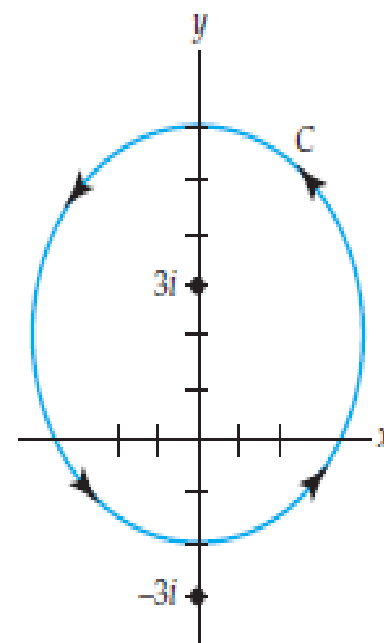
Evaluate  $\oint_C \frac{z}{z^2 + 9} dz$ , where  $C$  is the circle  $|z - 2i| = 4$ .

**Solution** By factoring the denominator as  $z^2 + 9 = (z - 3i)(z + 3i)$  we see that  $3i$  is the only point within the closed contour  $C$  at which the integrand fails to be analytic. See Figure . Then by rewriting the integrand as

$$\frac{z}{z^2 + 9} = \frac{\frac{z}{z + 3i}}{z - 3i},$$

we can identify  $f(z) = z/(z + 3i)$ . The function  $f$  is analytic at all points within and on the contour  $C$ . Hence, from Cauchy's integral formula (1) we have

$$\oint_C \frac{z}{z^2 + 9} dz = \int_C \frac{\frac{z}{z + 3i}}{z - 3i} dz = 2\pi i f(3i) = 2\pi i \frac{3i}{6i} = \pi i.$$



## Cauchy's Integral Formula

$$\oint_C \frac{e^z}{z-2} dz = 2\pi i e^z \Big|_{z=2} = 2\pi i e^2 = 46.4268i$$

for any contour enclosing  $z_0 = 2$  (since  $e^z$  is entire), and zero for any contour for which  $z_0 = 2$  lies outside (by Cauchy's integral theorem). 

## Cauchy's Integral Formula

$$\begin{aligned} \oint_C \frac{z^3 - 6}{2z - 1} dz &= \oint_C \frac{\frac{1}{2}z^3 - 3}{z - \frac{1}{2}i} dz \\ &= 2\pi i \left[ \frac{1}{2}z^3 - 3 \right]_{z=i/2} \\ &= \frac{\pi}{8} - 6\pi i \quad (z_0 = \frac{1}{2}i \text{ inside } C). \end{aligned}$$

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**Second Formula** We shall now build on Theorem by using it to prove that the values of the derivatives  $f^{(n)}(z_0)$ ,  $n = 1, 2, 3, \dots$  of an analytic function are also given by a integral formula. This second integral formula is similar to (1) and is known by the name **Cauchy's integral formula for derivatives**.

### Theorem Cauchy's Integral Formula for Derivatives

Suppose that  $f$  is analytic in a simply connected domain  $D$  and  $C$  is any simple closed contour lying entirely within  $D$ . Then for any point  $z_0$  within  $C$ ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (6)$$

### EXAMPLE 3 Using Cauchy's Integral Formula for Derivatives

Evaluate  $\oint_C \frac{z+1}{z^4+2iz^3} dz$ , where  $C$  is the circle  $|z|=1$ .

**Solution** Inspection of the integrand shows that it is not analytic at  $z=0$  and  $z=-2i$ , but only  $z=0$  lies within the closed contour. By writing the integrand as

$$\frac{z+1}{z^4+2iz^3} = \frac{\frac{z+1}{z+2i}}{z^3}$$

we can identify,  $z_0=0$ ,  $n=2$ , and  $f(z)=(z+1)/(z+2i)$ . The quotient rule gives  $f''(z)=(2-4i)/(z+2i)^3$  and so  $f''(0)=(2i-1)/4i$ . Hence from (6) we find

$$\oint_C \frac{z+1}{z^4+4z^3} dz = \frac{2\pi i}{2!} f''(0) = -\frac{\pi}{4} + \frac{\pi}{2}i.$$

### EXAMPLE 4 Using Cauchy's Integral Formula for Derivatives

Evaluate  $\int_C \frac{z^3 + 3}{z(z - i)^2} dz$ , where  $C$  is the figure-eight contour shown in Figure

**Solution** Although  $C$  is not a simple closed contour, we can think of it as the union of two simple closed contours  $C_1$  and  $C_2$  as indicated in Figure

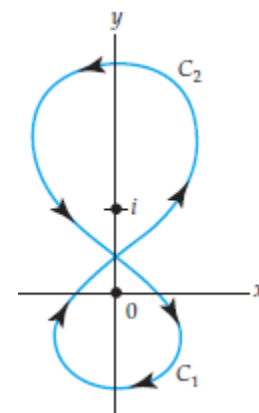
Since the arrows on  $C_1$  flow clockwise or in the negative direction, the opposite curve  $-C_1$  has positive orientation. Hence, we write

$$\begin{aligned}\int_C \frac{z^3 + 3}{z(z - i)^2} dz &= \int_{C_1} \frac{z^3 + 3}{z(z - i)^2} dz + \int_{C_2} \frac{z^3 + 3}{z(z - i)^2} dz \\ &= -\oint_{-C_1} \frac{\frac{z^3 + 3}{(z - i)^2}}{z} dz + \oint_{C_2} \frac{\frac{z^3 + 3}{z}}{(z - i)^2} dz = -I_1 + I_2,\end{aligned}$$

and we are in a position to use both formulas (1) and (6).

To evaluate  $I_1$  we identify  $z_0 = 0$ ,  $f(z) = (z^3 + 3)/(z - i)^2$ , and  $f(0) = -3$ . By (1) it follows that

$$I_1 = \oint_{-C_1} \frac{\frac{z^3 + 3}{(z - i)^2}}{z} dz = 2\pi i f(0) = 2\pi i(-3) = -6\pi i.$$





To evaluate  $I_2$  we now identify  $z_0 = i$ ,  $n = 1$ ,  $f(z) = (z^3 + 3)/z$ ,  $f'(z) = (2z^3 - 3)/z^2$ , and  $f'(i) = 3 + 2i$ . From (6) we obtain

$$I_2 = \oint_{C_2} \frac{\frac{z^3 + 3}{z}}{(z - i)^2} dz = \frac{2\pi i}{1!} f'(i) = 2\pi i(3 + 2i) = -4\pi + 6\pi i.$$

Finally, we get

$$\int_C \frac{z^3 + 3}{z(z - i)^2} dz = -I_1 + I_2 = 6\pi i + (-4\pi + 6\pi i) = -4\pi + 12\pi i.$$

## Some Consequences of the Integral Formulas

### Theorem Derivative of an Analytic Function Is Analytic

Suppose that  $f$  is analytic in a simply connected domain  $D$ . Then  $f$  possesses derivatives of all orders at every point  $z$  in  $D$ . The derivatives  $f', f'', f''', \dots$  are analytic functions in  $D$ .

### Theorem Cauchy's Inequality

Suppose that  $f$  is analytic in a simply connected domain  $D$  and  $C$  is a circle defined by  $|z - z_0| = r$  that lies entirely in  $D$ . If  $|f(z)| \leq M$  for all points  $z$  on  $C$ , then

$$\left| f^{(n)}(z_0) \right| \leq \frac{n!M}{r^n}. \quad (7)$$

The only bounded entire functions are constants.

If  $p(z)$  is a nonconstant polynomial, then the equation  $p(z) = 0$  has at least one root.

If  $f$  is continuous in a simply connected domain  $D$  and if  $\oint_C f(z) dz = 0$  for every closed contour  $C$  in  $D$ , then  $f$  is analytic in  $D$ .

Suppose that  $f$  is analytic and nonconstant on a closed region  $R$  bounded by a simple closed curve  $C$ . Then the modulus  $|f(z)|$  attains its maximum on  $C$ .

### EXAMPLE 5 Maximum Modulus

Find the maximum modulus of  $f(z) = 2z + 5i$  on the closed circular region defined by  $|z| \leq 2$ .

**Solution** We know that  $|z|^2 = z\bar{z}$ . By replacing the symbol  $z$  by  $2z + 5i$  we have

$$|2z + 5i|^2 = (2z + 5i)\overline{(2z + 5i)} = (2z + 5i)(2\bar{z} - 5i) = 4z\bar{z} - 10i(z - \bar{z}) + 25. \quad (8)$$

Since  $\bar{z} - z = 2i \operatorname{Im}(z)$ , and so (8) is

$$|2z + 5i|^2 = 4|z|^2 + 20 \operatorname{Im}(z) + 25. \quad (9)$$

Because  $f$  is a polynomial, it is analytic on the region defined by  $|z| \leq 2$ . By Theorem 1,  $\max_{|z| \leq 2} |2z + 5i|$  occurs on the boundary  $|z| = 2$ . Therefore, on  $|z| = 2$ , (9) yields

$$|2z + 5i| = \sqrt{41 + 20 \operatorname{Im}(z)}. \quad (10)$$

The last expression attains its maximum when  $\operatorname{Im}(z)$  attains its maximum on  $|z| = 2$ , namely, at the point  $z = 2i$ . Thus,  $\max_{|z| \leq 2} |2z + 5i| = \sqrt{81} = 9$ .

Evaluate the integral

$$\oint_C \frac{1}{z - z_0} dz,$$

where  $C$  is a circle centered at  $z_0$  and of any radius. The path is traced out once in the anticlockwise direction.

*Solution*

The circle can be parameterized by

$$z(t) = z_0 + re^{it}, \quad 0 \leq t \leq 2\pi,$$

where  $r$  is any positive real number. The contour integral becomes

$$\oint_C \frac{1}{z - z_0} dz = \int_0^{2\pi} \frac{1}{z(t) - z_0} \frac{dz(t)}{dt} dt = \int_0^{2\pi} \frac{ire^{it}}{re^{it}} dt = 2\pi i.$$

The value of the integral is independent of the radius  $r$ .

Evaluate the integral

$$(i) \int_C |z|^2 dz \text{ and } (ii) \int_C \frac{1}{z^2} dz,$$

where the contour  $C$  is

- (a) the line segment with initial point  $-1$  and final point  $i$ ;
- (b) the arc of the unit circle  $\operatorname{Im} z \geq 0$  with initial point  $-1$  and final point  $i$ .

Do the two results agree?

*Solution*

(i) Consider  $\int_C |z|^2 dz$ ,

(a) Parameterize the line segment by

$$z = -1 + (1 + i)t, \quad 0 \leq t \leq 1,$$

so that

$$|z|^2 = (-1 + t)^2 + t^2 \quad \text{and} \quad dz = (1 + i) dt.$$

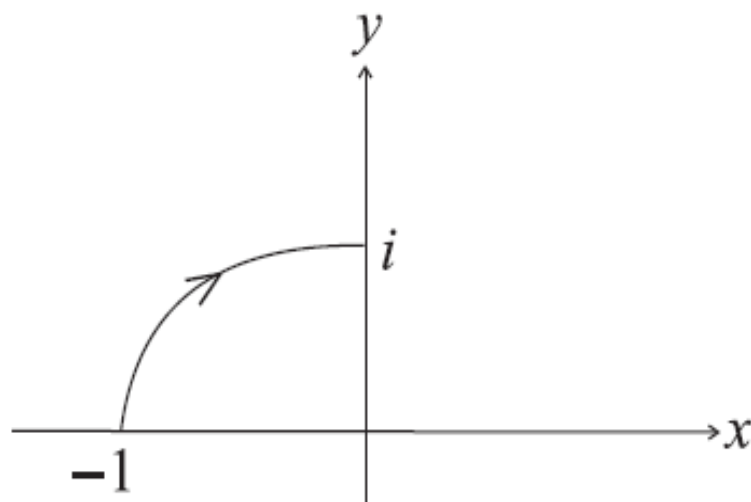
The value of the integral becomes

$$\int_C |z|^2 dz = \int_0^1 (2t^2 - 2t + 1)(1 + i) dt = \frac{2}{3}(1 + i).$$

(b) Along the unit circle,  $|z| = 1$  and  $z = e^{i\theta}$ ,  $dz = ie^{i\theta}d\theta$ . The initial point and the final point of the path correspond to  $\theta = \pi$  and  $\theta = \frac{\pi}{2}$ , respectively. The contour integral can be evaluated as

$$\int_C |z|^2 dz = \int_{\pi}^{\frac{\pi}{2}} ie^{i\theta} d\theta = e^{i\theta} \Big|_{\pi}^{\frac{\pi}{2}} = 1 + i.$$

The results in (a) and (b) do not agree. Hence, the value of this contour integral does depend on the path of integration.





(ii) Consider  $\int_C \frac{1}{z^2} dz$ .

(a) line segment from  $-1$  to  $i$

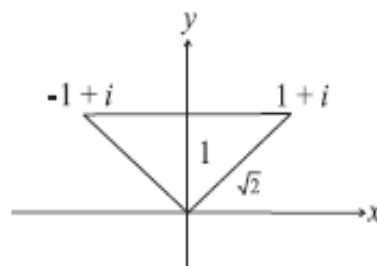
$$\int_C \frac{1}{z^2} dz = \int_0^1 \frac{1+i}{[-1+(1+i)t]^2} dt = -\frac{1}{-1+(1+i)t} \Big|_0^1 = -1 - \frac{1}{i} = -1 + i.$$

(b) subarc from  $-1$  to  $i$

$$\int_C \frac{1}{z^2} dz = \int_{\pi}^{\frac{\pi}{2}} \frac{1}{e^{2i\theta}} i e^{i\theta} d\theta = -e^{-i\theta} \Big|_{\pi}^{\frac{\pi}{2}} = -1 + i.$$

Show that

$$\left| \int_C \frac{1}{z^2} dz \right| \leq 2, \text{ where } C \text{ is the line segment joining } -1 + i \text{ and } 1 + i.$$



*Solution*

Along the contour  $C$ , we have  $z = x + i$ ,  $-1 \leq x \leq 1$ , so that  $1 \leq |z| \leq \sqrt{2}$ . Correspondingly,  $\frac{1}{2} \leq \frac{1}{|z|^2} \leq 1$ . Here,  $M = \max_{z \in C} \frac{1}{|z|^2} = 1$  and the arc length  $L = 2$ . We have

$$\left| \int_C \frac{1}{z^2} dz \right| \leq ML = 2.$$

Find an upper bound for  $\left| \int_{\Gamma} e^z / (z^2 + 1) dz \right|$ , where  $\Gamma$  is the circle  $|z| = 2$  traversed once in the counterclockwise direction.

*Solution*

The path of integration has length  $L = 4\pi$ . Next we seek an upper bound  $M$  for the function  $e^z / (z^2 + 1)$  when  $|z| = 2$ . Writing  $z = x + iy$ , we have

$$|e^z| = |e^{x+iy}| = e^x \leq e^2, \quad \text{for } |z| = \sqrt{x^2 + y^2} = 2,$$

and by the triangle inequality

$$|z^2 + 1| \geq |z|^2 - 1 = 4 - 1 = 3 \quad \text{for } |z| = 2.$$

Hence,  $|e^z / (z^2 + 1)| \leq e^2 / 3$  for  $|z| = 2$ , and so

$$\left| \int_{\Gamma} \frac{e^z}{z^2 + 1} dz \right| \leq \frac{e^2}{3} \cdot 4\pi.$$

If  $C$  is the curve  $y = x^3 - 3x^2 + 4x - 1$  joining points  $(1, 1)$  and  $(2, 3)$ , find the value of

$$\int_C (12z^2 - 4iz) dz.$$

*Method 1.* The integral is independent of the path joining  $(1, 1)$  and  $(2, 3)$ . Hence any path can be chosen. In particular, let us choose the straight line paths from  $(1, 1)$  to  $(2, 1)$  and then from  $(2, 1)$  to  $(2, 3)$ .

*Case 1* Along the path from  $(1, 1)$  to  $(2, 1)$ ,  $y = 1$ ,  $dy = 0$  so that  $z = x + iy = x + i$ ,  $dz = dx$ . Then the integral equals

$$\int_1^2 \{12(x + i)^2 - 4i(x + i)\} dx = \left\{ 4(x + i)^3 - 2i(x + i)^2 \right\} \Big|_1^2 = 20 + 30i.$$

*Case 2* Along the path from  $(2, 1)$  to  $(2, 3)$ ,  $x = 2$ ,  $dx = 0$  so that  $z = x + iy = 2 + iy$ ,  $dz = i dy$ . Then the integral equals

$$\int_1^3 \{12(2+iy)^2 - 4i(2+iy)\} i dy = \{4(2+iy)^3 - 2i(2+iy)^2\} \Big|_1^3 = -176 + 8i.$$

Then adding, the required value  $= (20 + 30i) + (-176 + 8i) = -156 + 38i$ .

*Method 2.* The given integral equals

$$\int_{1+i}^{2+3i} (12z^2 - 4iz) dz = (4z^3 - 2iz^2) \Big|_{1+i}^{2+3i} = -156 + 38i.$$

It is clear that Method 2 is easier.